United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Mid-Term Exam Trimester: Spring 2026
Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers
Total Marks: 30 Duration: 1 hour 30 minute(s)

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

	Question 1: Answer all the questions. (7.5 Marks) Transfer of bus control from processor to device takes **100 ns**. Transfer of bus control from device to processor takes **200 ns**.	**[CO1]**
a.	Shafi and Rafi are both transferring **1024 bytes** of data in **burst mode**. Shafi is using an I/O device with a data transfer rate of **x KB/µs (microseconds)**, while Rafi's I/O device has a data transfer rate of **2x KB/µs (microseconds)**. The total time required by Rafi is **50 nanoseconds** less than the time required by Shafi. **Find the value of x. (Show necessary mathematical calculations.)**	**[5]**
b.	While using parallel I/O in Programmed I/O mode, a user sets the command register to **95H**. The user then attempts to take input from pins **PA1, PA3, PA5, and PA6**. Will the system work correctly? Justify your answer.	**[2.5]**
	Question 2: Answer all the questions. (7.5 Marks)	**[CO1]**
a.	**AX = 8A32 H, BX = DFED H, CX = C321 H** Execute each of the following instructions **sequentially,** and determine the flags specified in each part. **Provide a brief explanation for your answer**. **i. ADD AX, BX ; Determine OF and PF.** **ii. SUB AX, CX ; Determine CF and AF.**	**[5]**
b.	During an 8086 microprocessor lecture, Higuruma says: **"The starting address of the code segment is 3AF28H."** Determine whether the statement is true or false. **Justify your answer.**	**[2.5]**
	Question 3: Answer all the questions. (7.5 Marks)	**[CO2]**
a.	**DS = 1E5B H, SS = 2F1D H, ES = 2BC2 H** I. Determine the **offset value** for the **physical address 2BCEC H** for each of the three segments, and justify whether the **offsets are valid or not**. II. Determine **how many physical memory address locations** overlap with the **Extra Segment (ES)** for each of the other two segments (**SS** and **DS**).	**[2 + 2.5]**

b.	Suppose there are one SRAM and two DRAMs ($DRAM_1$ and $DRAM_2$). The capacity ratio of **SRAM : $DRAM_1$: $DRAM_2$ is 1 : 2 : 4**. The **total combined memory capacity** of the three RAMs is **28 MB**. The data bus widths of **SRAM, $DRAM_1$, and $DRAM_2$** are **16 bits, 32 bits, and 64 bits, respectively**. Determine the **size** of their **address buses**.	**[3]**
Question 4: Answer all the questions. (7.5 Marks)		**[CO2]**
a.	You need to design a system where a microprocessor with an **18-bit address bus** and **16-bit data bus** is interfaced to an **18 KB RAM** system using **full decoding**, where the **second RAM** starts at address **2D800H**. Each RAM chip has an **11-bit address bus** and a **16-bit data bus**. **Draw a block diagram showing the microprocessor, RAM chips and full decoding logic**. Ensure the **second RAM starts at 2D800H**. **Provide the address mapping (first address and last address) for the first 3 RAM chips.**	**[4.5+ 3]**